

Introduction

Most financial fraud detection systems rely on **binary classifications**, **labeling transactions as fraud or legitimate**. While **effective for automation**, such decisions lack nuance and often **result in high false positives**. In contrast, real-world financial systems **require graded risk assessment that can rank transactions by risk and explain decisions**. This work proposes an explainable transaction risk scoring framework that assigns continuous risk scores instead of hard fraud flags.

Materials

Dataset

- Credit Card Transactions Dataset (Kaggle 2023)
- ~600,000 transactions
- 30 anonymized PCA-based features
- Balanced labels (50% fraud, 50% legitimate)

Balanced data enables stable training and focused analysis of risk scoring behavior and explainability, independent of class prevalence.

These calibrated probabilities are then transformed into **a continuous risk score** ($\text{Risk Score} = P(\text{fraud}) \times 100$), **converting binary predictions into a rankable risk measure**. SHAP (TreeExplainer) is applied to **attribute feature-level contributions**, providing **both global feature importance and per-transaction explanations**.

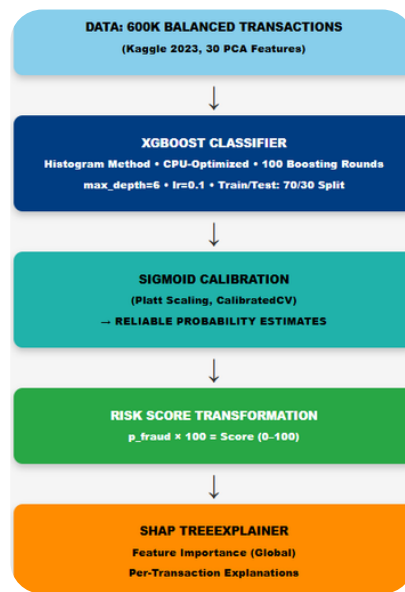


Fig 1. Pipeline Flowchart

Methodology

Modeling Pipeline

- XGBoost classifier (histogram method)
- 100 boosting rounds
- max_depth = 6, learning_rate = 0.1
- 70/30 train-test split

Raw model outputs are **calibrated** using **sigmoid (Platt) scaling** to ensure **predicted probabilities reflect relative model confidence**.

Results

- Fig. 2 shows **clear separation between legitimate and fraudulent transactions based on risk scores**.
- Fig.3 **confirms effective fraud classification** on balanced data via the confusion matrix.
- Fig.4 identifies the **most influential features driving fraud risk using SHAP importance**.
- Fig.5 shows the **directional impact of a key feature on predicted fraud risk**.
- Fig.6 demonstrates **stable model generalization through the ROC-AUC learning curve**.

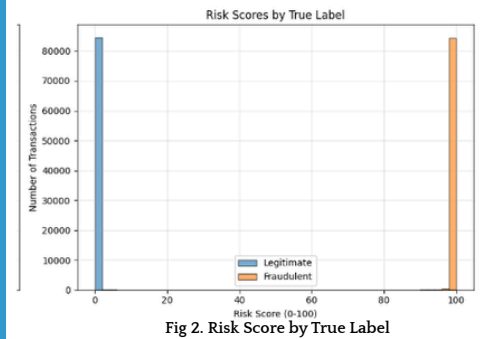


Fig 2. Risk Score by True Label

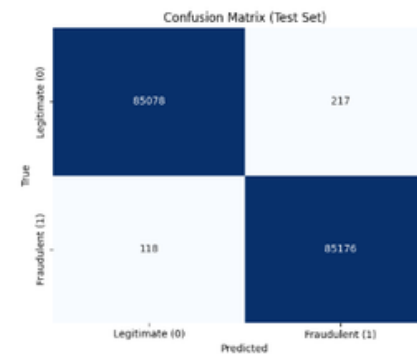


Fig 3. Confusion matrix confirms effective separation between fraudulent and legitimate transactions.

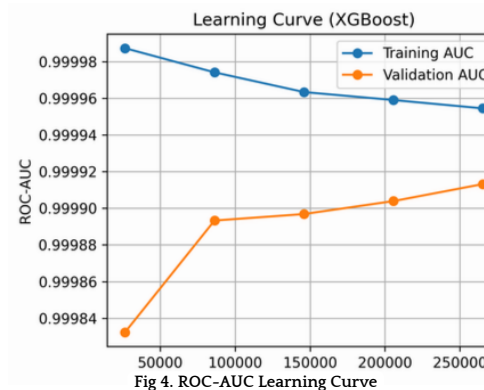


Fig 4. ROC-AUC Learning Curve

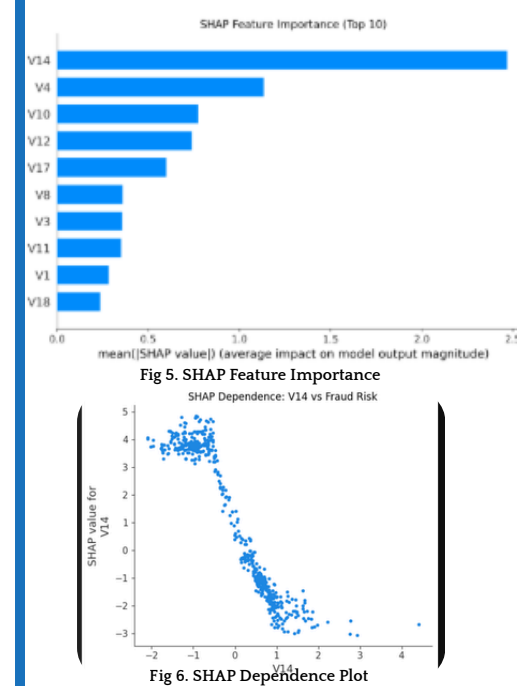


Fig 5. SHAP Feature Importance

Fig 6. SHAP Dependence Plot

Conclusion

This work shows how explainable risk scoring moves fraud detection beyond binary decisions. **Continuous risk scores provide clear separation between legitimate and fraudulent transactions**. SHAP-based explanations offer **transparency into model behaviour** enabling **confidence-based** financial risk assessment.

References

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